Systems of higher order equations

Solving systems of higher order equations will be a natural extension of what we learned previously about solving systems of two equations. In fact, we've already looked at several systems of three differential equations, like when we covered equal real Eigenvalues of multiplicity three.

Now we just want to formalize what we saw in those earlier examples, and cover systems of higher order equations more comprehensively. Our main goal here is just to understand how the number of solution combinations increase as the system gets larger.

Systems of two equations

We saw previously that we only had four possibilities for systems of two equations: two distinct real Eigenvalues, a repeated Eigenvalue producing one Eigenvector, a repeated Eigenvalue producing two Eigenvectors, or one pair of complex conjugate Eigenvalues.

1. 2 distinct real Eigenvalues

Eigenvalues	Eigenvectors	Solutions
λ_1	$\vec{k_1}$	$\vec{x_1} = \vec{k_1} e^{\lambda_1 t}$
λ_2	$\vec{k_2}$	$\vec{x_2} = \vec{k_2} e^{\lambda_2 t}$

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2. 2 complex conjugate Eigenvalues (λ_1 and λ_2 are conjugates, $\overrightarrow{k_1}$ and $\overrightarrow{k_2}$ are conjugates)

Eigenvalues	Eigenvectors	Solutions
$\lambda_1 = \alpha + \beta i$	$\overrightarrow{k_1}$	$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$
$\lambda_2 = \alpha - \beta i$	$\overrightarrow{k_2}$	$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$

3. 1 double real Eigenvalue giving 1 Eigenvector

Eigenvalues	Eigenvectors	Solutions
$\lambda_1 = \lambda_2$	$\overrightarrow{k_1}$	$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$
		$\overrightarrow{x_2} = \overrightarrow{k_1} t e^{\lambda_1 t} + \overrightarrow{p_1} e^{\lambda_1 t}$

$$(A - \lambda_1 I)\overrightarrow{p_1} = \overrightarrow{k_1}$$

4. 1 double real Eigenvalue giving 2 Eigenvectors

Eigenvalues	Eigenvectors	Solutions
$\lambda_1 = \lambda_2$	$\overrightarrow{k_1}$	$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$
	$\overrightarrow{k_2}$	$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_1 t}$

These $\overrightarrow{x_1}$ and $\overrightarrow{x_2}$ solutions were always the two linearly independent solutions that we needed in order to build the general solution to the system of two differential equations.

Systems of three equations

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As we might expect, to build the general solution to a system of three differential equations, we'll need three linearly independent solutions. Logically, to build the general solution to a system of n differential equations, we'll need n linearly independent solutions.

Before we look at all possible combinations of solutions to a system of three equations, we want to realize that there are restrictions on these combinations, based on the types of Eigenvalues we find.

For instance, if we have a repeated Eigenvalue in a system of three equations, or course it'll be repeated either two or three times (equal real Eigenvalues with multiplicity two, or equal real Eigenvalues with multiplicity three). Complex conjugate Eigenvalues always appear in pairs, so we'll either have exactly one pair of complex conjugate Eigenvalues, or no complex Eigenvalues at all. With these restrictions in mind, let's look at all possible solution combinations for a system of three differential equations. There are exactly seven possibilities.

1. 3 distinct real Eigenvalues

Eigenvalues	Eigenvectors	Solutions
λ_1	$\vec{k_1}$	$\vec{x_1} = \vec{k_1} e^{\lambda_1 t}$
λ_2	$\vec{k_2}$	$\vec{x_2} = \vec{k_2} e^{\lambda_2 t}$
λ_3	$\vec{k_3}$	$\vec{x_3} = \vec{k_3} e^{\lambda_3 t}$

2. 1 distinct real Eigenvalue, 2 complex conjugate Eigenvalues (λ_2 and λ_3 are conjugates, $\vec{k_2}$ and $\vec{k_3}$ are conjugates)

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Eigenvalues	Eigenvectors	Solutions
λ_1	$\overrightarrow{k_1}$	$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t}$
$\lambda_2 = \alpha + \beta i$	$\overrightarrow{k_2}$	$\overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t}$
$\lambda_3 = \alpha - \beta i$	$\overrightarrow{k_3}$	$\overrightarrow{x_3} = \overrightarrow{k_3}e^{\lambda_3 t}$

3. 1 distinct real Eigenvalue, 1 double real Eigenvalue giving 1 Eigenvector

Eigenvalues	Eigenvectors	Solutions
λ_1	$\overrightarrow{k_1}$	$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t}$
$\lambda_2 = \lambda_3$	$\overrightarrow{k_2}$	$\overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t}$
		$\overrightarrow{x_3} = \overrightarrow{k_2}te^{\lambda_2 t} + \overrightarrow{p_1}e^{\lambda_2 t}$

$$(A - \lambda_2 I)\overrightarrow{p_1} = \overrightarrow{k_2}$$

4. 1 distinct real Eigenvalue, 1 double real Eigenvalue giving 2 Eigenvectors

Eigenvalues	Eigenvectors	Solutions
λ_1	$\overrightarrow{k_1}$	$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t}$
$\lambda_2 = \lambda_3$	$\overrightarrow{k_2}, \overrightarrow{k_3}$	$\overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t}$
		$\overrightarrow{x_3} = \overrightarrow{k_3}e^{\lambda_3 t}$

5. 1 triple real Eigenvalue giving 1 Eigenvector

Eigenvalues	Eigenvectors	Solutions

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$\lambda_1 = \lambda_2 = \lambda_3$	$\vec{k_1}$	$\vec{x_1} = \vec{k_1}e^{\lambda_1 t}$
		$\vec{x_2} = \vec{k_1}te^{\lambda_1 t} + \vec{p_1}e^{\lambda_1 t}$
		$\vec{x_3} = \vec{k_1}\frac{t^2}{2}e^{\lambda_1 t} + \vec{p_1}te^{\lambda_1 t} + \vec{q_1}e^{\lambda_1 t}$
		$(A - \lambda_1 I)\vec{p_1} = \vec{k_1}$
		$(A - \lambda_1 I)\vec{q_1} = \vec{p_1}$

6. 1 triple real Eigenvalue giving 2 Eigenvectors

Eigenvalues	Eigenvectors	Solutions
$\lambda_1 = \lambda_2 = \lambda_3$	$\vec{k_1}$	$\vec{x_1} = \vec{k_1}e^{\lambda_1 t}$
	$\vec{k_2}$	$\vec{x_2} = \vec{k_2}e^{\lambda_2 t}$
		$\vec{x_3} = \vec{k_2}te^{\lambda_2 t} + \vec{p_1}e^{\lambda_2 t}$
		$(A - \lambda_2 I)\vec{p_1} = \vec{k_2}$

7. 1 triple real Eigenvalue giving 3 Eigenvectors

Eigenvalues	Eigenvectors	Solutions
$\lambda_1 = \lambda_2 = \lambda_3$	$\vec{k_1}$	$\vec{x_1} = \vec{k_1}e^{\lambda_1 t}$
	$\vec{k_2}$	$\vec{x_2} = \vec{k_2}e^{\lambda_1 t}$
	$\vec{k_3}$	$\vec{x_3} = \vec{k_3}e^{\lambda_1 t}$

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Fourth and higher order systems

So it starts to become clear that, the more differential equations we add to the system, the greater the number of possible Eigenvalue combinations, and the more complicated it'll become to build out the general solution.

For example, fourth order systems give us our first opportunity to find multiple pairs of complex conjugate roots. Systems of two or three differential equations simply aren't large enough to allow for this possibility. But once the system includes four or more equations, it becomes possible to find multiple pairs of complex conjugate Eigenvalues, and this obviously adds a little extra complication to finding the system's general solution.

That being said, let's do another example with a higher order equation to get more practice with the different solution combinations we've listed out in this lesson.

Example

Solve the system of differential equations.

$$x_1' = x_1 - x_3$$

$$x_2' = 2x_2 + x_3$$

$$x_3' = 2x_3$$

We'll need to start by finding the matrix $A - \lambda I$,

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$$A - \lambda I = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 0 & -1 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{bmatrix}$$

and then find its determinant $|A - \lambda I|$.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & -1 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)\begin{vmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} - (0)\begin{vmatrix} 0 & 1 \\ 0 & 2-\lambda \end{vmatrix} + (-1)\begin{vmatrix} 0 & 2-\lambda \\ 0 & 0 \end{vmatrix}$$

$$|A - \lambda I| = (1-\lambda)((2-\lambda)(2-\lambda) - (1)(0)) - ((0)(0) - (2-\lambda)(0))$$

$$|A - \lambda I| = (1-\lambda)(2-\lambda)(2-\lambda)$$

Solve the characteristic equation for the Eigenvalues.

$$(1-\lambda)(2-\lambda)(2-\lambda) = 0$$

$$\lambda = 1,\ 2,\ 2$$

We'll handle $\lambda_1 = 1$ first, starting with finding $A - 1I$.

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$$A - 1I = \begin{bmatrix} 1-1 & 0 & -1 \\ 0 & 2-1 & 1 \\ 0 & 0 & 2-1 \end{bmatrix}$$

$$A - 1I = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

This matrix is already simplified enough to turn it into a system of equations, so we get $k_3 = 0$ and

$$k_2 + k_3 = 0$$

$$k_2 = -k_3$$

$$k_2 = 0$$

So if we choose $k_1 = 1$, then the Eigenvector is

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

and therefore the solution vector is

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^{t}$$

Then for the repeated Eigenvalue $\lambda_2 = \lambda_3 = 2$, we find

$$A - 2I = \begin{bmatrix} 1-2 & 0 & -1 \\ 0 & 2-2 & 1 \\ 0 & 0 & 2-2 \end{bmatrix}$$

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$$A - 2I = \begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

If we turn this into a system of equations, we get $k_3 = 0$ and

$$-k_1 - k_3 = 0$$

$$k_1 = -k_3$$

$$k_1 = 0$$

So if we choose $k_2 = 1$, then the Eigenvector is

$$\overrightarrow{k_2} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

and therefore the solution vector is

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t}$$

Because we only find one Eigenvector for the two Eigenvalues $\lambda_2 = \lambda_3 = 2$, we have to use $\overrightarrow{k_2} = (0,1,0)$ to find a second solution.

$$(A - \lambda_2 I)\overrightarrow{p_1} = \overrightarrow{k_2}$$

$$\begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

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Expanding this matrix equation as a system of equations gives

$$-p_1 - p_3 = 0$$

$$p_3 = 1$$

Therefore, we get $p_1 = -1$. We choose $p_2 = 0$, so $\overrightarrow{p_1} = (-1,0,1)$. Then the second solution will be

$$\overrightarrow{x_3} = \overrightarrow{k_2} t e^{\lambda_2 t} + \overrightarrow{p_1} e^{\lambda_2 t}$$

$$\overrightarrow{x_3} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{2t}$$

and the general solution to the homogeneous system is

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} e^t + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} e^{2t} + c_3 \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} te^{2t} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} e^{2t} \right)$$

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